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AI ML

DAY 4 TASKS

1. Research Virtual Try on Ai Models

CP-VTON

Basic virtual try-on framework Uses clothing warping techniques Lightweight implementation

VITON-HD

Generates high-quality try-on images Better texture preservation Requires higher GPU resources

HR-VITON

Improved clothing fitting Better body alignment Suitable for realistic try-on systems Try On Diffusion Uses diffusion AI models Produces realistic outfit generation Slower inference speed

Conclusion

HR-VITON and Try On Diffusion provide better realism, while CP-VTON is easier to implement for beginners.

2. Methods Tested

Methods	Speed	Accuracy	Real-time support
Media pipe selfie Segmentation	fast	medium	yes
Deeplab v3	slow	high	Limited
body pix	medium	medium	yes

Observation

DeepLabV3 produced the best segmentation quality, while MediaPipe worked best for real-time applications.

3. Clothing recommendation logic

To create a basic clothing recommendation system based on body type and clothing style.

Logic flow

Input User Image

↓

Detect Body Shape

↓

Analyze Clothing Style

↓

Match Outfit Dataset

↓

Generate Recommendation

```
Enter occasion (casual/formal/party/sports): formal
```

```
Recommended Clothes:
```

- Oversized T-Shirts
- Layered Jackets
- Baggy Jeans
- Blazer
- Formal Shoes
- Stylish Handbag

```
PS C:\Users\91630>
```

4. Media pipe vs open pose

Feature	Media pipe	open pose
Technology used	lightweight ml pipelines	deep learning CNN
speed	very fast	moderate
Accuracy	good	excellen
Gpu requirement	low	high
Multi person detection	limited	strong
Out put	basic body landmarks	detailed body key points

5.Improve Pose Detection Accuracy

To improve human pose detection performance for different body positions and lighting conditions.

Work done

Tested pose detection using MediaPipe
Increased detection confidence threshold
Improved image preprocessing
Tested multiple poses and distances
Reduced missing body keypoints

